

CIVICTRACKER — Complete Backend & Technical Architecture

API Lifecycles • Vision AI Pipeline • Spatial GIS Geometry • Database Models • Security

1. BACKEND OVERVIEW & TECH STACK

API Runtime:	Next.js 15 Serverless Edge API Routes (/api/v1/ai/...)	— Stateless autoscaling 10 - 1M requests
Database:	Firebase Firestore (Realtime DB) + PostgreSQL DDL	— Sub-second real-time sync & audit trails
AI Vision:	Google Gemini 2.0 / 1.5 Flash Vision API + YOLOv8	— Dual-stage intake validation & repair audit
Spatial GIS:	Ray-Casting Point-in-Polygon + OSM Nominatim	— Automated ward polygon routing & distance
Auth & RBAC:	Firebase Auth with Role-Based Access Control	— Verified Citizen vs. Municipal Admin permissions

2. CORE API ROUTES & REQUEST LIFECYCLES

POST /api/v1/ai/gemini-analyze (Stage-1 Intake Vision) 1. Client sends { image: base64, category_hint?: string }. 2. Gemini Flash Vision evaluates image as a certified municipal civil engineer. 3. Validates defect type (POTHOLE, GARBAGE, WATER_LEAK, OPEN_MANHOLE), computes severity (1-5), and confidence score. 4. Rejects memes, selfies, and non-defect photos before docket creation.
POST /api/v1/ai/gemini-verify-resolution (Stage-2 Repair Auditor) 1. Receives { before_image, after_image, defect_type }. 2. Sends multi-part payload to Gemini Vision to compare Before vs. After structural integrity. 3. Checks if pothole is leveled, garbage cleared, or leak welded. 4. Returns binary YES (Resolved -> status flip) or NO (Rework Required -> blocks closure).

3. DATABASE ARCHITECTURE (Firestore & SQL)

Collection [civic_issues]: id, complaint_number (CTR-2026-...), category, title, description, status, severity (1-5), lat/lng, zone_id, zone_name, department, reported_at, deadline_at (+15d), resolved_at, image_url, resolution_image_url, upvotes
Collection [civic_users]: uid, email, displayName, photoURL, role (citizen admin), permanentCivicScore (100-1000), quarterlyPoints (Q1-Q4), tier (Platinum/Gold/Silver/Bronze)
Collection [issue_status_history]: id, issue_id, previous_status, new_status, changed_by, change_note, timestamp (Immutable audit trail)

4. SPATIAL GIS ALGORITHM & SLA ENGINE

Point-in-Polygon (Ray-Casting GIS) Mathematical algorithm in zoneMatcher.ts that casts a horizontal ray from citizen GPS coordinates across municipal ward polygon vertices. Dispatches dockets directly to PWD, Sanitation, or Water Board with 0ms manual routing latency.
15-Day Statutory SLA & Escalation Calculates deadline_at = reported_at + 15 days. Real-time cron and query auditors flag overdue tickets publicly on the Accountability Leaderboard, notifying the Municipal Commissioner. 500 upvotes compresses SLA to 48h emergency priority.
Civic Sense Scoring & Quarterly Reset +50 pts reporting, +100 pts resolution, +30 pts validation, +10 pts upvote. Permanent score (100-1000) preserved indefinitely; competition points reset every 3 months (Q1-Q4) to award official Government Merit Certificates.

5. SECURITY & PERFORMANCE ARCHITECTURE

- Client-Side Compression: HTML5 Canvas compresses high-res camera photos to ~150KB, enabling instant uploads over 2G/3G.
- Tamper-Proof Audit Trail: All status changes are logged immutably in `issue_status_history` with timestamp and author.
- Role-Based Access (RBAC): Department workers require authenticated role tokens; citizens can only modify their own dockets.
- Real-Time Scalability: Firestore reactive snapshots deliver instant live leaderboard updates to thousands of connected clients.